

EDUCATION

- **University of Maryland, Baltimore County (UMBC)**
Ph.D. in Computer Science
 - Research: Large Language Models, Knowledge Graphs, HPC, NLP
 - **University of Maryland, Baltimore County (UMBC)**
Master of Science in Computer Science; GPA: 3.74
 - **University of Maryland, Baltimore County (UMBC)**
Graduate Certificate in College Teaching and Learning Science
 - **University of Maryland, Baltimore County (UMBC)**
Bachelor of Science in Computer Science; GPA: 3.756, Magna Cum Laude
 - **University of Maryland, Baltimore County (UMBC)**
Bachelor of Arts in Political Science; GPA: 3.756, Magna Cum Laude
 - Honors: Pi Sigma Alpha – Political Science Honor Society
 - Student Government Association Finance Board Representative

Baltimore, MD
Dec. 2025

Baltimore, MD
May 2023

Baltimore, MD
May 2023

Baltimore, MD
Jan. 2021

Baltimore, MD
Jan. 2021
- **Carroll Community College (CCC)**
Associate of Applied Science: Cybersecurity
 - Student Government Association Senator

Westminster, MD
May 2018

EXPERIENCE

- **Los Alamos National Laboratory (LANL)**
Scientist II
 - **Research Areas:** Designed and implemented scalable data pipelines, statistical models, and AI/ML algorithms to extract actionable insights from large, complex, and heterogeneous datasets.
 - **Collaboration:** Collaborated with interdisciplinary teams of scientists and engineers to translate research innovations and PhD-level computer science expertise into operational analytics, decision-support tools, and high-impact technical deliverables.

Los Alamos, NM
Oct. 2026 – present
- **Los Alamos National Laboratory (LANL)**
Graduate Research Assistant
 - **Research Areas:** Conducted research in Large Language Models, Knowledge Graphs, Tensor Factorization, High-performance Computing, and Natural Language Processing.
 - **Collaboration:** Collaborated with Theoretical Chemistry and Cybersecurity teams on interdisciplinary projects.

Los Alamos, NM
Oct. 2022 – Oct. 2026
- **University of Maryland, Baltimore County (UMBC)**
Adjunct Lecturer, Department of Computer Science
 - **Course Instruction:** Taught CMSC 201, 210, 310, 341, and 462. Focused on programming, data structures, and data science in Python and C++.

Baltimore, MD
Aug. 2022 – Jul. 2023
- **University of Maryland, Baltimore County (UMBC)**
Graduate Teaching Assistant, Department of Computer Science
 - **Instructional Support:** Supported CMSC 304, 421, 426, and 473/673 through grading, office hours, guest lectures, and course development.

Baltimore, MD
Jan. 2021 – Jan. 2023
- **University of Maryland, Baltimore County (UMBC)**
Undergraduate Teaching Fellow & Assistant, Department of Computer Science
 - **Peer Instruction:** Led discussions, graded assignments, and mentored students in CMSC 201 & CMSC 341.

Baltimore, MD
Aug. 2019 – Dec. 2020

PUBLICATIONS

1. **Barron, R. C.**, Eren, M. E., Stanev, V., Matuszek, C., Alexandrov, B. S. *Topic Modeling and Link-Prediction for Material Property Discovery*. In *DocEng '25: 25th ACM Symposium on Document Engineering*, Sep. 2–5, 2025, University of Nottingham, UK. 4 pages.
2. Bhattarai, M., **Barron, R.**, Eren, M. E., Vu, M., Grantcharov, V., Boureima, I., Stanev, V., Matuszek, C., Valtchinov, V., Rasmussen, K. *HEAL: Hierarchical Embedding Alignment Loss for Improved Retrieval and Representation Learning*. In *ICLR '25 Workshop on Scaling Self-Improving Foundation Models without Human Supervision*, Apr. 21, 2025, Singapore. 10 pages.
3. **Barron, R.**, Eren, M. E., Truong, D. P., Matuszek, C., Wendelberger, J., Dorn, M. F., Alexandrov, B. S. *Matrix Factorization for Inferring Associations and Missing Links*. *arXiv preprint arXiv:2503.04680*, 2025.
4. **Barron, R. C.**, Eren, M. E., Serafimova, O. M., Matuszek, C., Alexandrov, B. S. *Bridging Legal Knowledge and AI*. In *ICAIL '25: 20th International Conference on Artificial Intelligence and Law*, Jun. 16–20, 2025, Chicago, IL, USA. 10 pages.
5. **Barron, R. C.**, Grantcharov, V., Wanna, S., Eren, M. E., Bhattarai, M., Solovyev, N., Tompkins, G., Nicholas, C., Rasmussen, K., Matuszek, C. *Domain-Specific Retrieval-Augmented Generation Using Vector Stores, Knowledge Graphs, and Tensor Factorization*. In *ICMLA '24: 23rd IEEE Conference on Machine Learning and Applications—Special Session on ML for NLP*, Dec. 18–20, 2024, Miami, FL, USA. 8 pages.
6. **Barron, R.**, Eren, M. E., Bhattarai, M., Boureima, I., Matuszek, C., Alexandrov, B. S. *Binary Bleed: Fast Distributed and Parallel Method for Automatic Model Selection*. In *IEEE High Performance Extreme Computing Conference (HPEC)*, pp. 1–8, 2024.
7. Eren, M. E., **Barron, R.** *Catch'em All: Classification of Rare, Prominent, and Novel Malware Families*. In *12th International Symposium on Digital Forensics and Security (ISDFS)*, pp. 1–6, 2024. IEEE.
8. **Barron, R.**, Eren, M. E. *Cyber-Security Knowledge Graph Generation by Hierarchical Nonnegative Matrix Factorization*. In *12th International Symposium on Digital Forensics and Security (ISDFS)*, pp. 1–6, 2024. IEEE.
9. Wanna, S., Solovyev, N., **Barron, R.**, Eren, M. E., Bhattarai, M., Rasmussen, K. Ø, Alexandrov, B. S. *Topictag: Automatic Annotation of NMF Topic Models Using Chain of Thought and Prompt Tuning with LLMs*. In *Proceedings of the ACM Symposium on Document Engineering*, pp. 1–4, 2024.
10. Bhattarai, M., Kaymak, M. C., **Barron, R. C.**, Nebgen, B., Rasmussen, K. Ø, Alexandrov, B. S. *Defense of Supervised and Unsupervised Machine Learning Models*. Technical Report, 2023.
11. Higgins, P., **Barron, R.**, Engel, D., Matuszek, C. *A Comparative Analysis of VR-Based and Real-World Human-Robot Collaboration for Small-Scale Joining*. In *IEEE Conference on Virtual Reality and 3D User Interfaces Workshop (VRW)*, pp. 845–846, 2023.
12. Higgins, P., **Barron, R.**, Lukin, S., Engel, D., Matuszek, C. *A Collaborative Building Task in VR vs. Reality*. In *International Symposium on Experimental Robotics*, pp. 11–21, 2023. Springer Nature Switzerland Cham.
13. Solovyev, N., **Barron, R.**, Bhattarai, M., Eren, M. E., Rasmussen, K. Ø, Alexandrov, B. S. *Interactive Distillation of Large Single-Topic Corpora of Scientific Papers*. In *2023 International Conference on Machine Learning and Applications (ICMLA)*, pp. 1000–1005, 2023. IEEE.
14. Bhattarai, M., Kaymak, M. C., **Barron, R.**, Nebgen, B., Rasmussen, K. Ø, Alexandrov, B. S. *Robust Adversarial Defense by Tensor Factorization*. In *2023 International Conference on Machine Learning and Applications (ICMLA)*, pp. 308–315, 2023. IEEE.
15. Higgins, P., **Barron, R.**, Engel, D., Matuszek, C. *Lessons from a Small-Scale Robot Joining Experiment in VR*. 2023.

16. Higgins, P., **Barron, R.**, Matuszek, C. *Head Pose as a Proxy for Gaze in Virtual Reality*. In *5th International Workshop on Virtual, Augmented, and Mixed Reality for HRI*, 2022.
17. Higgins, P., **Barron, R.**, Matuszek, C. *Head Pose for Object Deixis in VR-Based Human-Robot Interaction*. In *2022 31st IEEE International Conference on Robot and Human Interactive Communication (RO-MAN)*, pp. 610–617, 2022. IEEE.
18. Kebe, G. Y., Higgins, P., Jenkins, P., Darvish, K., Sachdeva, R., **Barron, R.**, Winder, J., Engel, D., Raff, E., Ferraro, F. *A Spoken Language Dataset of Descriptions for Speech-Based Grounded Language Learning*. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2021.

CERTIFICATIONS

- **Apple Ads Certification** — Apple, 2024
- **Neo4j-backed Chatbot Certification** — Neo4j, November 2023
- **Neo4j & Large Language Models Fundamentals** — Neo4j, November 2023
- **Applying the QM Rubric** — Quality Matters, October 2021
- **Social/Behavioral Research Course** — CITI Program, May 2021

SKILLS

- **Programming Languages:** Python, C, C++, Go, Fortran, UNIX shell
- **Software & Tools:** Git, Docker, L^AT_EX, Neo4j, Conda, LLM Prompting
- **Technical Areas:** Tensor Factorization, Knowledge Graphs, Machine Learning, Natural Language Processing, Robotics
- **Other Skills:** Academic research, teaching, training, consultation